

WEEK -2

PRACTICAL

Name :- Abhay Tripathi

Roll no :- 2520030353

Sub :- DBSE&DBD

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\abhay>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookflow_db1;
Query OK, 1 row affected (0.01 sec)

mysql> USE bookflow_db1;
Database changed
mysql> CREATE TABLE Books (
-> book_id INT PRIMARY KEY,
-> title VARCHAR(100) NOT NULL,
-> isbn VARCHAR(20) UNIQUE,
-> published_year INT CHECK (published_year < 2027)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
-> (1, 'The Great Gatsby', '9780743273565', 1925),
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn                | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565      | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084      | 1960           |
+-----+-----+-----+-----+
```

```
C:\Windows\system32\cmd.e: X + v
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Members (
-> member_id INT PRIMARY KEY,
-> full_name VARCHAR(100),
-> email VARCHAR(100) UNIQUE
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Members (member_id, full_name, email) VALUES
-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email                |
+-----+-----+-----+
| 101       | John Smith | john.smith@email.com |
| 102       | Emma Wilson | emma.wilson@email.com |
| 103       | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Loans (
-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
C:\Windows\system32\cmd.e: X + v

mysql> CREATE TABLE Loans (
  -> loan_id INT PRIMARY KEY,
  -> member_id INT,
  -> book_id INT,
  -> loan_date DATE,
  -> FOREIGN KEY (member_id) REFERENCES Members(member_id),
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
  -> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
  -> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
  -> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
  -> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
  -> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
  -> FROM Loans l
  -> INNER JOIN Members m ON l.member_id = m.member_id
  -> INNER JOIN Books b ON l.book_id = b.book_id;
```

```

mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;

```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

10 rows in set (0.00 sec)

```

mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;

```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.00 sec)

```

mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );

```

Query OK, 0 rows affected (0.07 sec)

```

C:\Windows\system32\cmd.e: X + v

mysql> CREATE TABLE Donation_History (
  -> donation_id INT PRIMARY KEY,
  -> book_id INT,
  -> donor_name VARCHAR(100),
  -> donation_date DATE,
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
  -> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
  -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935'
  ->
  -> \c
mysql> USE bookflow_db1;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db1 |
+-----+
| books                   |
| donation_history        |
| loans                   |
| members                 |

```

```
C:\Windows\system32\cmd.e:  X  +  v

mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db1 |
+-----+
| books                    |
| donation_history         |
| loans                    |
| members                  |
+-----+
4 rows in set (0.01 sec)

mysql> DESCRIBE Books;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id    | int       | NO   | PRI | NULL    |       |
| title      | varchar(100) | NO   |     | NULL    |       |
| isbn       | varchar(20) | YES  | UNI | NULL    |       |
| published_year | int       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> DESCRIBE Members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id  | int       | NO   | PRI | NULL    |       |
| full_name  | varchar(100) | YES  |     | NULL    |       |
| email      | varchar(100) | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> DESCRIBE Loans;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| loan_id    | int       | NO   | PRI | NULL    |       |
| member_id  | int       | YES  | MUL | NULL    |       |
| book_id    | int       | YES  | MUL | NULL    |       |
| loan_date  | date      | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
```

```
C:\Windows\system32\cmd.e: X + v

mysql> DESCRIBE Donation_History;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| donation_id | int       | NO   | PRI | NULL    |       |
| book_id     | int       | YES  | MUL | NULL    |       |
| donor_name  | varchar(100) | YES  |     | NULL    |       |
| donation_date | date      | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby    | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
| 4       | Animal Farm         | 9780451526342 | 1945           |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email                |
+-----+-----+-----+
| 101       | John Smith    | john.smith@email.com |
| 102       | Emma Wilson   | emma.wilson@email.com |
| 103       | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1       | 101       | 1       | 2025-01-05 |
| 2       | 102       | 2       | 2025-01-08 |
| 3       | 103       | 3       | 2025-01-10 |
+-----+-----+-----+-----+
```

```
C:\Windows\system32\cmd.e: X + v

|      101 | John Smith | john.smith@email.com |
|      102 | Emma Wilson | emma.wilson@email.com |
|      103 | Michael Brown | michael.brown@email.com |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Loans;
+-----+
| loan_id | member_id | book_id | loan_date |
+-----+
|      1 |      101 |      1 | 2025-01-05 |
|      2 |      102 |      2 | 2025-01-08 |
|      3 |      103 |      3 | 2025-01-10 |
|      4 |      101 |      2 | 2025-02-01 |
|      5 |      102 |      1 | 2025-02-05 |
|      6 |      103 |      2 | 2025-02-12 |
|      7 |      101 |      3 | 2025-03-01 |
|      8 |      102 |      3 | 2025-03-07 |
|      9 |      103 |      1 | 2025-03-15 |
|     10 |      101 |      1 | 2025-04-01 |
+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM Donation_History;
+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+
|          1 |      4 | Raj Kumar | 2026-07-30 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM Donation_History;|
```